HOSPITAL RESOURCE OPTIMIZATION AND DEMAND FORECASTING

Project Id: 25-26J-226

Individual Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology
(Specializing in Information Technology)

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

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I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Signature of supervisor: Date: 27/08/2025

ACKNOWLEDGEMENT

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ABSTRACT

The effective management of outpatient appointment schedules is essential to ensuring timely access to healthcare, optimal utilization of medical staff, and overall operational efficiency in hospitals. However, patient “no-shows,” where individuals miss scheduled appointments without prior cancellation, remain a pervasive challenge worldwide—especially in resource-limited healthcare systems. This issue leads to underutilized clinical capacity, extended patient waiting times, and increased financial strain on healthcare providers.

This research introduces a predictive modeling framework aimed at forecasting the likelihood of patient no-shows and enabling proactive scheduling interventions. Drawing on historical appointment records, patient demographics, attendance histories, and contextual factors such as day-of-week patterns or seasonal trends, the study leverages supervised machine learning algorithms—including logistic regression, random forest, and gradient boosting models—to uncover underlying behavioral patterns. The resulting predictions inform targeted interventions, such as personalized reminders, selective overbooking, and dynamic slot allocation, to minimize the impact of missed appointments.

While automated scheduling systems are common in modern healthcare, the integration of predictive analytics into these processes remains underexplored in many developing countries. This study emphasizes adaptability to low-resource contexts, ensuring that the proposed solution can be implemented using existing hospital information systems without requiring extensive infrastructure upgrades.

The significance of this research lies in its potential to transform reactive scheduling into a proactive, data-driven process. By accurately identifying high-risk appointments, hospitals can reduce idle doctor time, improve patient flow, and enhance continuity of care. Beyond its immediate operational benefits, this framework also contributes to the growing field of AI-driven healthcare optimization, offering a scalable and practical model for improving efficiency in emerging healthcare systems.

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LIST OF ABBREVIATION

Table 1 : Abbreviations

Abbreviation	**Full Form**
AI	Artificial Intelligence
EHR	Electronic Health Records
LSTM	Long Short-Term Memory
SSL	Secure Sockets Layer
HTTPS	Hypertext Transfer Protocol Secure
UI	User Interface
NN	Neural Network
XGBoost	Extreme Gradient Boosting
F1-score	F-measure (Harmonic Mean of Precision and Recall)

1. INTRODUCTION

1.1. Background & Literature

In the dynamic realm of healthcare delivery, where human well-being and operational precision converge, the outpatient appointment system serves as a vital link between patients and medical professionals. Yet, this system is frequently hindered by a persistent challenge—**patient no-shows**—where scheduled appointments are missed without prior notice. These absences ripple through the hospital's ecosystem, leading to underutilized clinical resources, disrupted schedules, extended waiting times, and diminished quality of care. While digital hospital management systems have introduced notable efficiency to scheduling, a promising frontier remains underexplored—the integration of **predictive analytics** to anticipate and mitigate no-shows before they occur.

A historical reflection on hospital appointment management reveals a journey from manual, paper-based registers to modern electronic scheduling platforms. While this transformation has improved accessibility and record-keeping, it has largely remained reactive, addressing attendance issues after they disrupt operations. The growing availability of historical appointment data—enriched with patient demographics, attendance histories, and temporal patterns—now presents an opportunity to move beyond reaction toward **proactive decision-making**. This shift mirrors the evolution in other industries, where data-driven forecasting has become a cornerstone of efficiency.

The application of **machine learning techniques** to predict patient attendance represents the next leap in this transformation. Models such as **Logistic Regression, Random Forest, Gradient Boosting (XGBoost), and Neural Networks** offer the capacity to uncover hidden correlations between patient behavior and no-show tendencies. These models can integrate variables like appointment day-of-week, lead time between booking and consultation, past attendance history, seasonal influences, and even external factors such as public holidays or adverse weather conditions. Literature in this field underscores the tangible benefits of predictive modeling in healthcare scheduling. Studies have demonstrated reductions in no-show rates by up to 30% through targeted interventions informed by model outputs, such as automated reminders or selective overbooking strategies. Moreover, research highlights the practicality of implementing these models in resource-limited settings by relying on readily available structured data rather than costly, advanced infrastructure

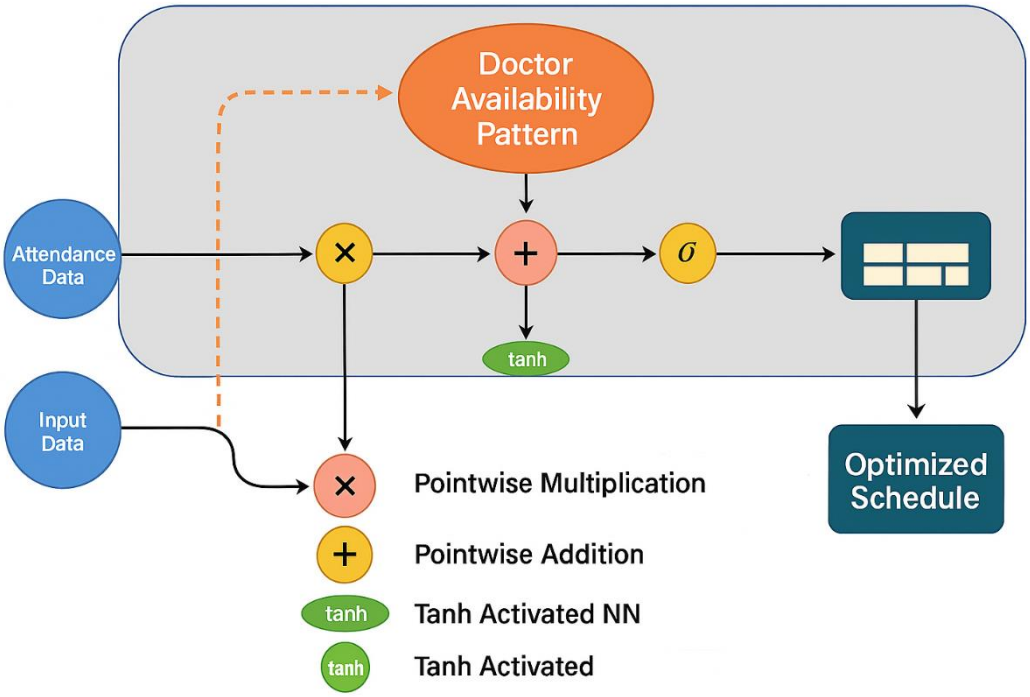


Figure 1 : LSTM Networks Detailed Explanation

However, this research extends beyond the mathematical and computational dimensions. It aspires to bridge **technological precision** with the **human-centered ethos** of healthcare. By embedding predictive intelligence into appointment systems, hospitals can preserve the integrity of patient care while optimizing scarce medical resources. This convergence of **artificial intelligence and compassionate healthcare delivery** creates a narrative where every appointment slot becomes a well-utilized opportunity—reducing waste, enhancing efficiency, and ensuring that both patients and providers benefit from a more reliable, informed scheduling process.

1.2. Research Gap

While healthcare institutions worldwide have begun exploring predictive analytics to address operational inefficiencies, the application of such technologies in Sri Lanka's outpatient appointment systems remains limited. Existing scheduling approaches in local hospitals are largely reactive, focusing on managing appointments once they are missed rather than preventing no-shows in the first place. This leads to wasted consultation slots, underutilized medical staff, and increased patient wait times — inefficiencies that could be mitigated with proactive prediction and intervention.

Globally, no-show prediction models using machine learning have been shown to improve hospital resource management and patient flow. However, these models are often developed in contexts with robust electronic health record (EHR) infrastructures, advanced patient tracking systems, and abundant computational resources — conditions that do not fully align with the realities of many Sri Lankan hospitals. Consequently, there is a lack of localized models that can operate effectively within the constraints of developing healthcare systems, using readily available data without requiring extensive technological overhauls.

Furthermore, there is minimal integration between predictive no-show models and hospital decision-making workflows in Sri Lanka. Even where data is available, it is rarely leveraged for dynamic scheduling adjustments, targeted reminder systems, or proactive overbooking strategies. This gap in both research and practice presents an opportunity to develop a **context-specific, resource-efficient, and interpretable** predictive solution. Such a system would not only address operational challenges but also improve patient access to care, thereby aligning with national healthcare objectives.

1.3. Research Problem

The research problem at hand focuses on enhancing outpatient clinic efficiency through the strategic integration of machine learning techniques. Specifically, it seeks to address the question of how predictive models can be effectively utilized to forecast the likelihood of patient no-shows and, in turn, optimize appointment slot allocation for upcoming clinic schedules. This prediction will be based on historical appointment records, patient attendance patterns, and related contextual variables available within existing hospital systems.

In the evolving landscape of healthcare delivery — where the balance between limited resources and growing patient demand is critical — the predictive potential of machine learning in mitigating appointment no-shows remains largely underutilized in Sri Lanka. While existing hospital management systems can record and manage appointments, they are primarily reactive, dealing with missed slots after they occur rather than proactively preventing them. This leaves a significant gap in the ability to forecast attendance behavior and dynamically adjust scheduling to ensure optimal use of clinical resources.

This research problem revolves around leveraging historical appointment data, including prior attendance records, appointment characteristics, and patient-specific factors, to identify patterns that can inform predictive models. By combining these patterns with hospital operational workflows, the study aims to develop an intelligent, context-specific tool capable of flagging high-risk appointments in advance. This will enable healthcare staff to take preventive measures such as targeted reminders or strategic overbooking, thereby reducing idle time and improving patient flow.

The complexity of this problem is multifaceted. It involves designing and training supervised machine learning models — such as Random Forest and Gradient Boosting — that can handle diverse predictors and imbalanced attendance data while providing interpretable results for clinical decision-making. Additionally, it requires careful integration with hospital workflows to ensure that predictions translate into actionable interventions. Addressing these challenges will not only enhance scheduling efficiency but also contribute to better patient care continuity, reduced waiting times, and improved overall healthcare service quality.

2. OBJECTIVES

2.1. Main Objectives

The primary objective of this research component is to design and implement a predictive analytics system that enhances the efficiency and reliability of doctor-patient scheduling in large-scale hospital environments. In many healthcare institutions, patient appointments are often scheduled without sufficient consideration of historical doctor attendance patterns, leading to frequent mismatches, appointment cancellations, and delays in treatment. These issues not only disrupt hospital workflows but also negatively impact patient satisfaction and trust in the healthcare system.

This component aims to address these challenges by developing a data-driven solution that analyzes historical doctor attendance records to identify recurring patterns and trends. For example, certain doctors may frequently take leave on specific weekdays, during public holidays, or around seasonal events. By recognizing these patterns, the system can forecast potential absences and proactively adjust appointment schedules to ensure that patients are matched with available doctors.

The proposed system will consist of several key stages:

- **Data Collection and Preprocessing**: Gathering historical attendance data from hospital databases, cleaning the data to remove inconsistencies, and structuring it for analysis.
- **Feature Engineering**: Extracting relevant features such as day-of-week, holiday proximity, department-specific trends, and individual doctor behavior to improve model accuracy.
- **Model Development**: Training machine learning models—such as Long Short-Term Memory (LSTM) networks, decision trees, or ensemble methods—to predict doctor availability on future dates. These models will be evaluated using metrics like accuracy, precision, and recall to ensure reliability.
- **Scheduling Optimization**: Integrating the predictions into a scheduling engine that recommends optimal appointment slots based on expected doctor availability. This engine will also flag high-risk days and suggest alternative scheduling options to minimize disruptions.

In addition to improving operational efficiency, this system introduces a layer of intelligence and adaptability to hospital resource planning. By shifting from reactive to proactive scheduling, hospitals can reduce patient wait times, improve resource utilization, and enhance the overall quality of care. Furthermore, the use of explainable

AI techniques will ensure that the predictions and recommendations are transparent and interpretable by hospital staff, fostering trust and ease of adoption.

Ultimately, this component contributes to the broader goal of building a smart hospital management system that leverages artificial intelligence to solve real-world problems in healthcare delivery. It demonstrates how predictive modeling can be applied not only to clinical outcomes but also to administrative processes, creating a more responsive and patient-centric healthcare environment.

2.2. Specific Objectives

There are five specific objectives that must be reached in order to achieve the overall objective described above.

1. **Collect and preprocess historical doctor attendance and scheduling data**
    - Gather detailed attendance records, appointment logs, and calendar data from hospital databases.
    - Clean and standardize the data to remove inconsistencies, missing entries, and formatting issues.
    - Engineer relevant features such as day-of-week patterns, seasonal leave trends, public holidays, and department-specific behaviors.
    - Organize the data into structured formats suitable for time-series and classification modeling.

2. **Design and implement predictive models to forecast doctor availability**
    - Select appropriate machine learning algorithms such as LSTM, decision trees, or ensemble methods based on the nature of the data.
    - Train models using historical attendance data to identify patterns and predict future absences.
    - Validate model performance using metrics like accuracy, precision, recall, and F1-score.
    - Integrate contextual factors (e.g., holidays, hospital events) to improve prediction accuracy.

3. **Develop a scheduling optimization engine based on predicted availability**
    - Create a rule-based or AI-driven scheduling system that uses forecasted availability to suggest optimal appointment slots.
    - Implement logic to flag high-risk days and recommend rescheduling or alternative doctor assignments.
    - Ensure the system supports real-time updates and can adapt to last-minute changes in availability.

4. **Evaluate and fine-tune the predictive scheduling system**
    - Test the system using real-world hospital scenarios and historical appointment outcomes.
    - Compare predicted availability with actual attendance to assess reliability.
    - Fine-tune model parameters and scheduling rules to improve performance and usability.
    - Collect feedback from hospital staff to ensure the system is practical and user-friendly.

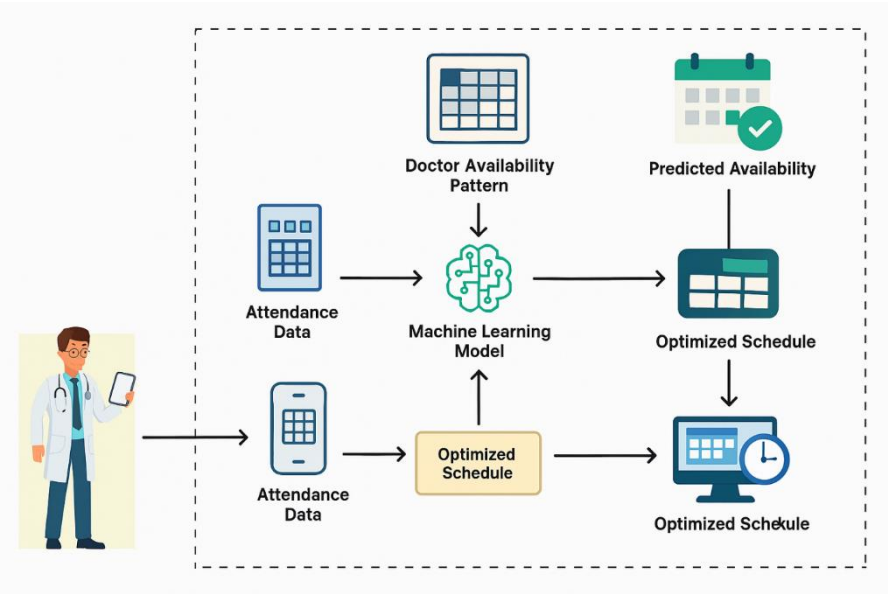
5. **Ensure transparency and explainability of predictions**
    - Incorporate explainable AI techniques to make model decisions interpretable by healthcare professionals.
    - Provide visualizations and summaries that show why certain predictions or scheduling suggestions were made.
    - Build trust among users by making the system's logic clear and accessible.

3. METHODOLOGY

This research adopts a structured methodology to develop a predictive and intelligent system for optimizing doctor-patient scheduling in large-scale hospitals. The process begins with the collection of historical data, including doctor attendance records, patient appointment logs, hospital calendars, and public holidays. This data is then cleaned and preprocessed to remove inconsistencies, fill in missing values, and standardize formats. Important features such as day-of-week trends, seasonal leave patterns, department-specific behaviors, and holiday-related absences are extracted to enhance the predictive power of the model. Once the data is prepared, machine learning algorithms—such as Long Short-Term Memory (LSTM) networks and decision tree-based classifiers—are trained to forecast doctor availability on specific days. These models are evaluated using performance metrics like accuracy, precision, recall, and F1-score to ensure reliability. The predictions generated by the model are then integrated into a scheduling optimization engine that recommends suitable appointment slots, flags high-risk days for potential absences, and suggests alternative arrangements to minimize mismatches between patients and doctors. The system is tested against traditional manual scheduling methods to assess improvements in appointment success rates and overall efficiency. Visual tools such as graphs and comparison charts are used to present the results clearly. To ensure long-term effectiveness, the system is designed to continuously learn from new data and adapt to changing hospital dynamics, making it a sustainable and intelligent solution for improving healthcare service delivery.

3.1. The System Overview Diagram

By collecting and refining historical data and integrating data-driven inputs, we create a robust dataset. Through the application of machine learning algorithms, we build predictive models capable of anticipating tea production needs. The models undergo rigorous evaluation, finetuning, and comparison with traditional manual predictions, enabling us to validate their effectiveness. This comprehensive process ensures that the resulting predictions are informed, accurate, and capable of guiding strategic decisions in the dynamic landscape of tea production. Our methodology's continuous improvement loop underscores our commitment to refining the models' performance over time, facilitating smarter and more efficient tea production practices.



3.1.1. Summary of Technologies

Summary of technologies, techniques, architectures and algorithms used is shown in the table below.

Table 2 : Technologies

Description	Tools and Technologies
Frontend Technologies	React Native
Data Analysis	Python (Pandas, NumPy)
Backend Technologies	NodeJS, Python
Machine learning	scikit-learn, TensorFlow
Database for store data	MongoDB
Hosting the API	Heroku
Weather API	Google Cloud Healthcare API
Version Controlling	Git
Team connectivity	Teams and WhatsApp

3.2. Use Case Diagram

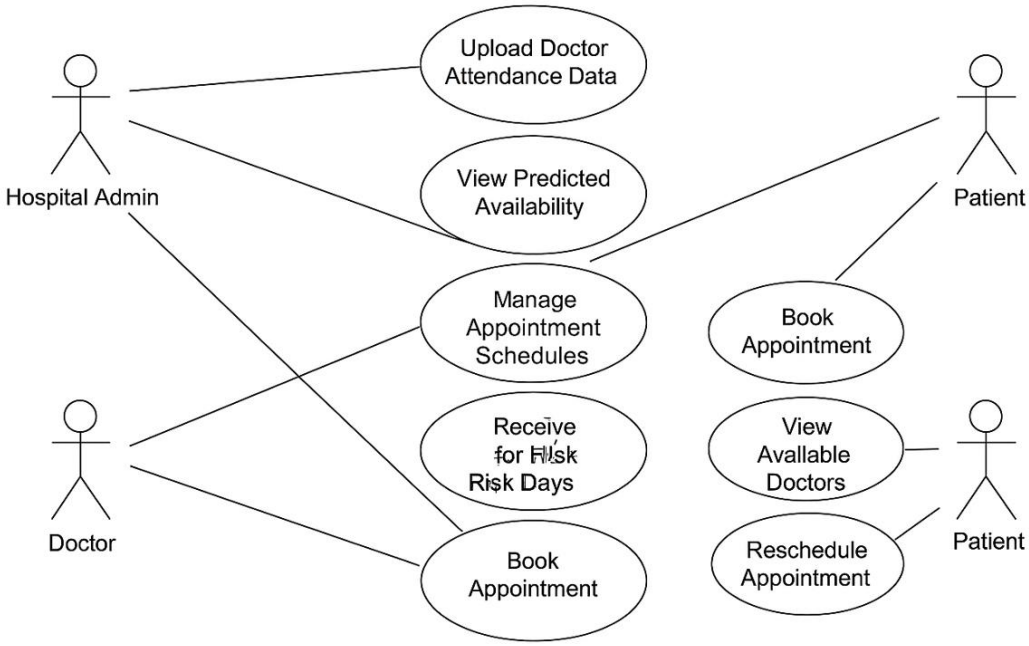


Figure 3 : Use case Diagram

3.3. Sequence Diagram

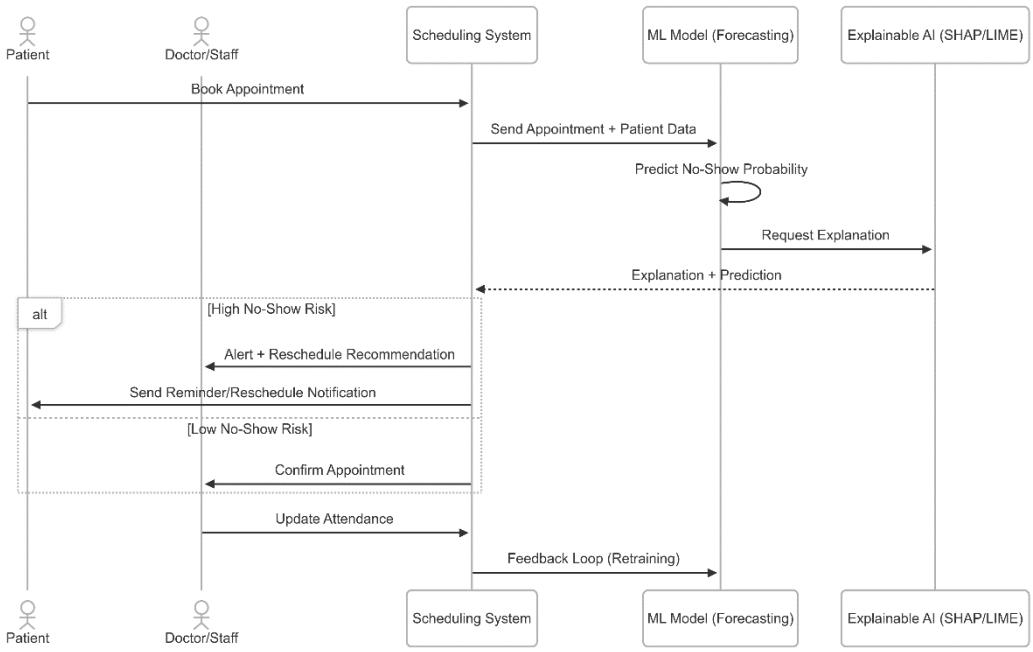


Figure 4 : Sequence Diagram

3.4. Class Diagram

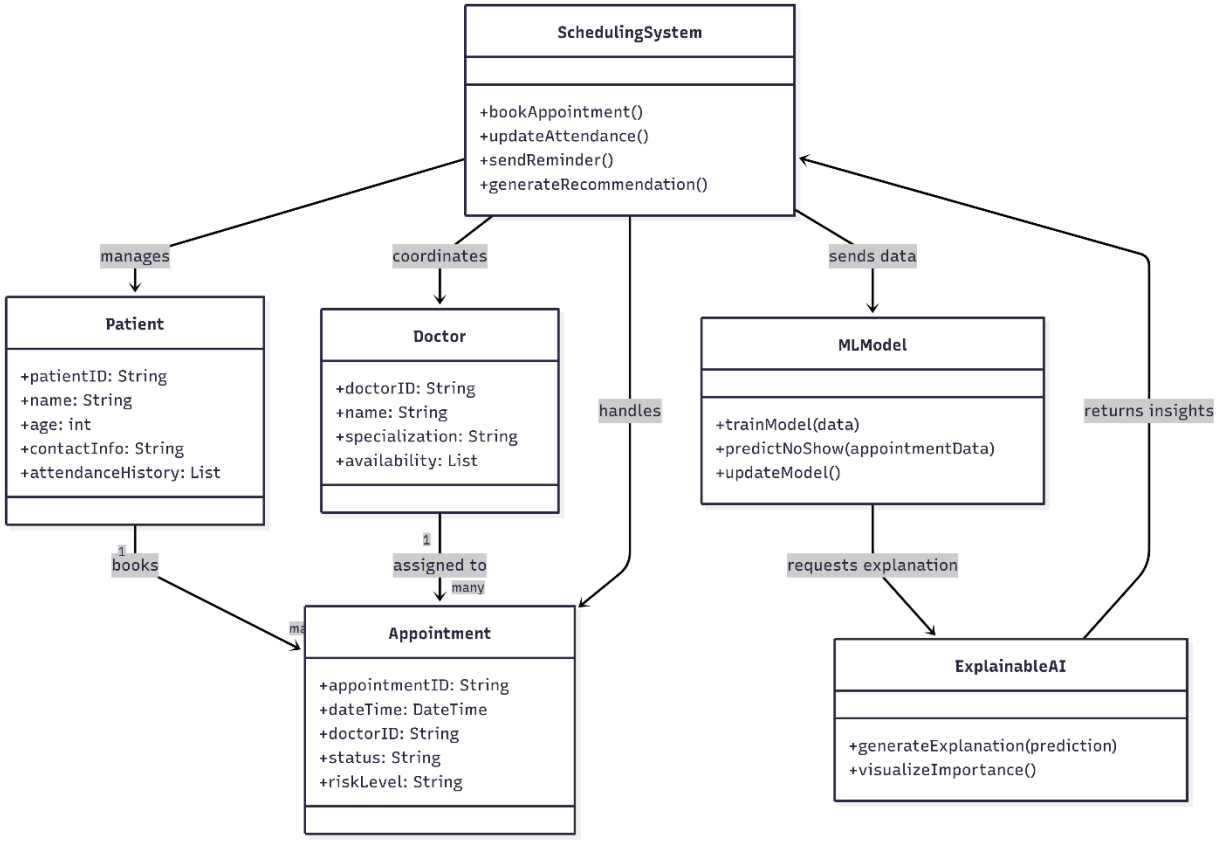


Figure 5 : Class Diagram

4. REQUIREMENTS

The requirements for this system are categorized into user, functional, and system requirements. These requirements ensure that the system addresses the needs of hospital staff while being technically feasible and sustainable.

4.1. User Requirements

- The system should be able to collect historical doctor attendance and appointment data from multiple hospital sources.
- The system should clean and preprocess the collected data to remove errors, inconsistencies, and missing values.
- The system should identify and extract relevant features such as day-of-week patterns, seasonal leave trends, and holiday-related absences.
- The system should provide a selection of machine learning algorithms suitable for time-series forecasting and classification.
- The system should train selected models using the preprocessed data and optimize them for accuracy.
- The system should validate model performance using separate validation datasets to detect overfitting or underfitting.
- The system should generate predictions of doctor availability and integrate them into a scheduling engine.
- The system should provide recommendations for optimal appointment slots based on predicted availability.
- The system should allow hospital staff to view, interpret, and trust predictions through explainable AI features.
- The system should support continuous learning by updating predictions based on new attendance data.

4.2. Functional Requirements

1. Performance
   - The system should generate predictions and scheduling recommendations within a reasonable timeframe.
   - It should support multiple concurrent users without performance degradation.
2. Accuracy
   - Doctor availability predictions should closely match actual attendance records.
   - Scheduling suggestions should reduce mismatches and cancellations.
3. Reliability
   - The system should be consistently available to support daily hospital operations.
   - It should recover gracefully from unexpected failures without data loss.
4. Scalability
   - The system should handle increasing volumes of attendance and scheduling data.
   - It should support the addition of new features and models without compromising performance.

4.3. System Requirements

1. Hardware Requirements
   - Minimum and recommended specifications for servers or computers running the system.
   - Sufficient storage capacity for historical data, model parameters, and scheduling logs.
2. Software Requirements
   - Compatible operating systems (e.g., Windows, Linux).
   - Required libraries and frameworks (e.g., TensorFlow, PyTorch, scikit-learn).

3. Database Requirements

- A robust database system (e.g., PostgreSQL, MongoDB) for storing raw data, processed features, and model outputs.
- Scalable storage for growing datasets and model artifacts.

4. Network and Connectivity

- Reliable internet connectivity for accessing external data sources and updates.
- Network infrastructure to support real-time data exchange and multi-user access.

5. User Interface Requirements

- A user-friendly interface for hospital staff to interact with predictions and scheduling tools.
- Compatibility with major web browsers and devices (desktop, tablet, mobile).

6. Security Requirements

- Secure data transmission using encryption protocols (e.g., HTTPS, SSL).
- Role-based access control with user authentication and authorization.

7. Performance Requirements

- Fast response times for generating predictions and scheduling suggestions.
- Capacity to handle a defined number of concurrent users or requests.

8. Scalability Requirements

- Ability to scale with increasing data volume and user demand.
- Support for cloud-based deployment to enhance flexibility and scalability.

5. MARKETABILITY

The proposed AI-powered hospital resource optimization system exhibits strong market potential due to its direct alignment with critical challenges faced by healthcare institutions globally. Hospitals, especially large-scale ones, frequently struggle with resource inefficiencies, including bed shortages, medicine stockouts, and mismatched doctor-patient scheduling—issues that this system is specifically designed to address.

Key market drivers include:

- **Rising healthcare demand** due to aging populations, seasonal illness surges, and emergency events.
- **Operational inefficiencies** in current hospital systems that rely on static, reactive planning.
- **Growing interest in AI and predictive analytics** within the healthcare sector to improve decision-making and reduce costs.

Target customers include public and private hospitals, healthcare networks, and government health departments. These stakeholders are increasingly seeking smart solutions that can enhance patient care while optimizing operational costs.

The system's **modular architecture**—featuring components for bed forecasting, illness trend detection, doctor scheduling, and medicine inventory management—makes it highly adaptable across various hospital sizes and specialties. This scalability ensures that the solution can be customized for different healthcare environments, from urban hospitals to rural clinics.

Moreover, the use of **explainable AI** builds trusts among healthcare professionals, a critical factor for adoption in clinical settings. Integration with existing hospital information systems and electronic health records (EHRs) is streamlined through the use of proven technologies such as time-series models, NLP, and predictive analytics.

In the long term, the system has the potential to evolve into a **standardized hospital resource management platform**, enabling data-driven collaboration across departments and institutions. This positions it not only as a commercial product but also as a transformative tool for improving healthcare delivery and resilience.

6. GANTT CHART

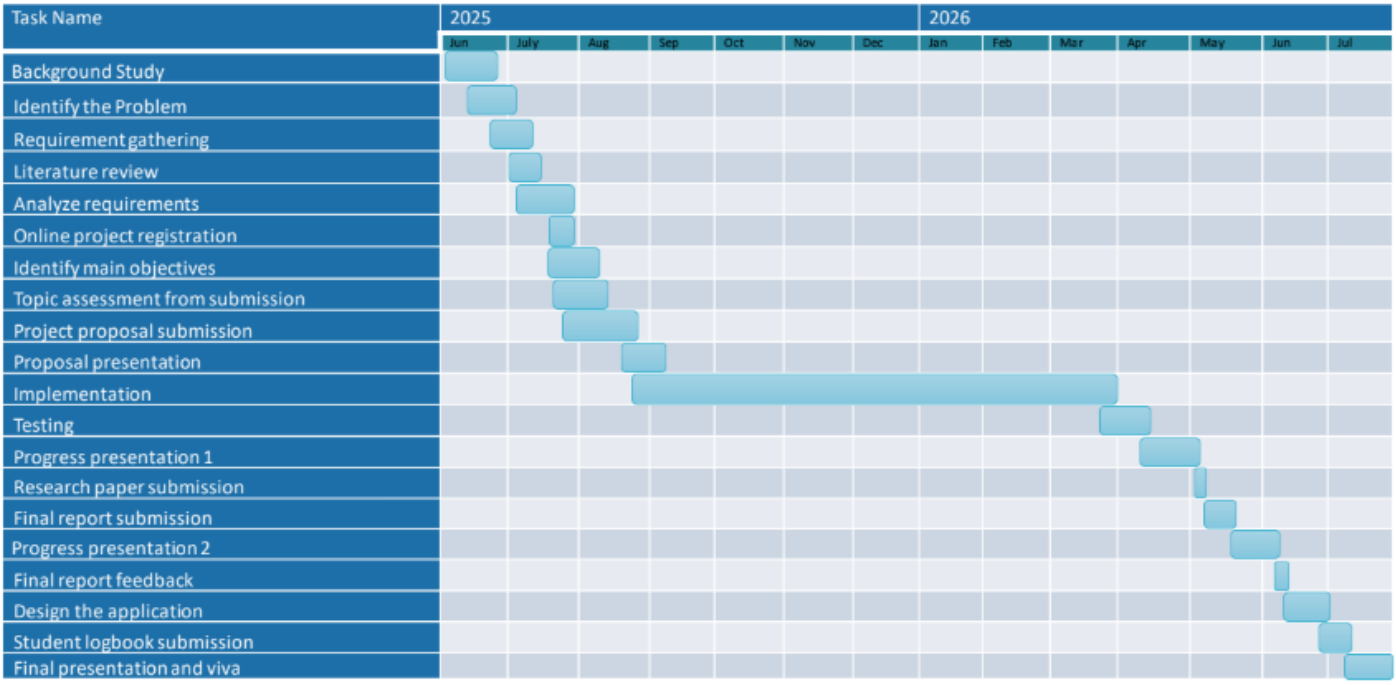


Figure 6 : Gantt Chart

7. WORK BREAKDOWN STRUCTURE

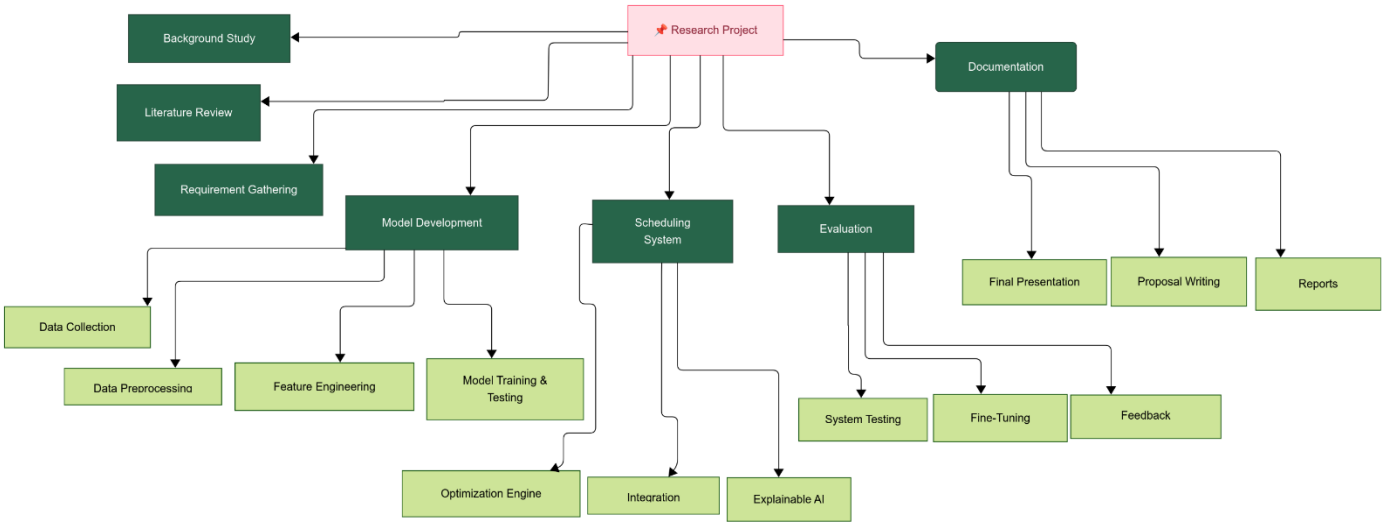


Figure 7 : Work Breakdown Structure

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